

# Blackmagic Streaming Ethernet Protocol



Blackmagic Streaming Encoder HD  
Blackmagic Streaming Encoder 4K

# Developer Information

## Blackmagic Streaming Encoder Ethernet Protocol

### v1.2

#### Protocol Details

##### Overview

The Blackmagic Streaming Encoder Ethernet Protocol is a line-oriented, text-based protocol to control a Streaming Encoder.

Lines from the Streaming Encoder server will be separated by an ASCII LF sequence.

Messages from the user may be separated by LF or CR LF.

##### Connection

The Streaming Encoder server is accessed by connecting to TCP port 9977 on a Streaming Encoder.

##### Connection Response

Upon connection, the Streaming Encoder server sends a dump of the device's state.

The Streaming Encoder server sends information in blocks, with each having an identifying header followed by a colon. A block spans multiple lines and is terminated by a blank line. Each line in the protocol is terminated by a newline character.

Following the header, a block contains either a single multi-line string or a sequence of key/value pairs arranged one per line. The first full-colon on a line is used to delimit the key and the value. A value may be a comma separated list. In this case, values in the list must have the “,” and “\” characters escaped with a “\” character.

To be resilient to future protocol changes, clients should ignore blocks they do not recognize, up to the trailing blank line. Within recognized blocks, clients should ignore keys they do not recognize.

##### Legend

|             |                  |
|-------------|------------------|
| ↵           | End of line      |
| ...         | and so on        |
| Orange Text | Client Generated |
| Grey Text   | Server Generated |

The protocol preamble block is always the first block sent by the Streaming Encoder server:

```
PROTOCOL PREAMBLE:↵
Version: 1.2↵
↵
```

The version field indicates the protocol version. When the protocol is changed in a compatible way, the minor version number will be updated. If incompatible changes are made, the major version number will be updated.

The initial status dump is concluded by the end prelude block:

```
END PRELUDE:↵
↵
```

## Status Updates

When any device parameter is changed on the Streaming Encoder server by any client, such as the Blackmagic Streaming Setup utility, the Streaming Encoder server resends the applicable status block, containing only the items that have changed. Status updates can also occur due to external changes such as tethering to a smartphone or when a streaming service is disconnected.

For example, if the input video mode is set to Auto, the following block will be sent:

```
STREAM SETTINGS:↵
Video Mode: Auto↵
↵
```

## Requesting Changes

To change one or more parameters in a block, the client should send the appropriate block header in the same form the Streaming Encoder server sends, followed by the key/value pairs to be changed. For example, to change the input video format to 1080p59.94, the user should send the following block:

```
STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
↵
```

The block must be terminated by a blank line. On receipt of a blank line, the Streaming Encoder server will either acknowledge the request by responding:

```
ACK↵
↵
```

or if unable to parse the block responding with:

```
NACK↵
↵
```

After a positive response, the client should expect to see a status update from the Streaming Encoder server showing the status change. This is likely to be the same as the command that was sent, sometimes followed by other blocks providing data specific to the change.

```
STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
↵
```

If the Streaming Encoder server does not understand a key in the requested block, the key will be ignored. If an invalid value is provided for a known key, then the request is ignored and the Streaming Encoder will respond with the existing value for the key. In both cases the Streaming Encoder server will still respond with an ACK.

```
STREAM SETTINGS:↵
Video Mode: UnsupportedMode↵
↵
ACK↵
↵
STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
↵
```

## Requesting a Status Dump

The user may request that the Streaming Encoder server resend the complete state of any status block by sending the block header, followed by a blank line. In the following example, the user requests the Streaming Encoder server resend the stream settings:

```
STREAM SETTINGS:↵
↵
ACK↵
↵
STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
...
↵
```

## Protocol Blocks

### Identity Block

The identity block contains information to identify the connected Streaming Encoder.

#### Block Syntax

The following example shows the Identity Block for a Blackmagic Streaming Encoder HD.

```
IDENTITY:↵
Model: Blackmagic Streaming Encoder HD↵
Label: Blackmagic Streaming Encoder HD↵
Unique ID: 00112233445566778899AABBCCDDEEFF↵
↵
```

#### Parameters

| Key       | Read/Write | Description                              | Valid Values   |
|-----------|------------|------------------------------------------|----------------|
| Model     | Read only  | The Streaming Encoder model name         | String         |
| Label     | Read/Write | A display name for the Streaming Encoder | String         |
| Unique ID | Read only  | A device specific unique identifier      | Hexadecimal ID |

### Changing Device Label

A device label to identify the Streaming Encoder can be changed by sending an identity block with label key.

```
IDENTITY:↵
Label: My Streaming Encoder↵
↵
ACK↵
↵
IDENTITY:↵
Label: My Streaming Encoder↵
↵
```

## Version Block

The version block contains hardware and software version information for the connected Streaming Encoder.

### Block Syntax

```
VERSION:↵
Product ID: BE73↵
Hardware Version: 0100↵
Software Version: 0123ABCD↵
Software Release: 3.5↵
↵
```

### Parameters

| Key              | Read/Write | Description                                    | Valid Values        |
|------------------|------------|------------------------------------------------|---------------------|
| Product ID       | Read only  | The Streaming Encoder product ID               | Hexadecimal ID      |
| Hardware Version | Read only  | The Streaming Encoder hardware version         | Hexadecimal version |
| Software Version | Read only  | The Streaming Encoder software version         | Hexadecimal version |
| Software Release | Read only  | The Streaming Encoder software release version | Version Number      |

## Network Blocks

The network block contains the TCP/IP networking configuration for the connected Streaming Encoder.

### Block Syntax

This example shows the output for a connected Streaming Encoder. It displays 2 networking interfaces - the Gigabit Ethernet interface and option for a tethered smartphone.

The network settings prefixed with Current show the active TCP/IP settings, and are read-only. The Current settings reflect either the DHCP or Static configuration, depending on the Dynamic IP flag.

```
NETWORK:↵
Interface Count: 2↵
Default Interface: 0↵
↵
NETWORK INTERFACE 0:↵
Name: Ethernet↵
Priority: 1↵
MAC Address: 00:11:22:33:44:55↵
Dynamic IP: true↵
Current Addresses: 192.168.1.10/255.255.255.0↵
Current Gateway: 192.168.1.1↵
Current DNS Servers: 192.168.1.1, 8.8.8.8, 8.8.4.4↵
Static Addresses: 10.0.0.2/255.255.255.0↵
Static Gateway: 10.0.0.1↵
Static DNS Servers: 8.8.8.8, 8.8.4.4↵
↵
```

```

NETWORK INTERFACE 1:↵
Name: USBEthernet↵
Priority: 0↵
MAC Address: 00:00:00:00:00:00↵
Dynamic IP: true↵
Current Addresses: 0.0.0.0/255.255.0.0↵
Current Gateway: 0.0.0.0↵
Current DNS Servers: ↵
Static Addresses: 10.0.0.2/255.255.255.0↵
Static Gateway: 10.0.0.1↵
Static DNS Servers: 8.8.8.8, 8.8.4.4↵
↵

```

## Parameters

### Network Block

| Key               | Read/Write | Description                                                            | Valid Values |
|-------------------|------------|------------------------------------------------------------------------|--------------|
| Interface Count   | Read only  | The number of networking interfaces supported by the Streaming Encoder | Integer      |
| Default Interface | Read only  | The default networking interface                                       | Integer      |

### Network Interface Block

| Key                 | Read/Write | Description                                                                                                                                | Valid Values                                               |
|---------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Name                | Read only  | The name of the networking interface                                                                                                       | String                                                     |
| Priority            | Read/Write | The priority of the network interface. When multiple network interfaces are available, the high priority interface will become the default | Unsigned integer. The higher number is the higher priority |
| MAC Address         | Read Only  | MAC address of the networking interface                                                                                                    | IEEE 802 MAC address                                       |
| Dynamic IP          | Read/Write | Selects DHCP or Static IP configuration                                                                                                    | true - DHCP enabled<br>false - Static IP                   |
| Current Addresses   | Read Only  | The current IP address and Subnet mask                                                                                                     | {IPv4 address}/{Subnet Mask}                               |
| Current Gateway     | Read Only  | The current IP gateway address                                                                                                             | IPv4 address                                               |
| Current DNS Servers | Read only  | The IP addresses of the current DNS servers                                                                                                | Comma separated list of IPv4 addresses                     |
| Static Addresses    | Read/Write | Status IP address and subnet mask when DHCP disabled                                                                                       | {IPv4 address}/{Subnet Mask}                               |
| Static Gateway      | Read/Write | Static gateway address when DHCP disabled                                                                                                  | IPv4 address                                               |
| Static DNS Servers  | Read/Write | The IP addresses of the static DNS servers                                                                                                 | Comma separated list of IPv4 addresses                     |

## Changing Networking Settings

The network can be configured to use either DHCP or a static configuration. To enable DHCP on Network Interface 0:

```
NETWORK INTERFACE 0:↵
Dynamic IP: true↵
↵
ACK↵
↵
NETWORK INTERFACE 0:↵
Dynamic IP: true↵
↵
```

To set a fixed IP address, supply all static parameters:

```
NETWORK INTERFACE 0:↵
Dynamic IP: false↵
Static Addresses: 192.168.1.2/255.255.255.0↵
Static Gateway: 192.168.1.1↵
Static DNS Servers: 8.8.8.8, 8.8.4.4↵
↵
ACK↵
↵
NETWORK INTERFACE 0:↵
Dynamic IP: false↵
Static Addresses: 192.168.1.2/255.255.255.0↵
Static Gateway: 192.168.1.1↵
Static DNS Servers: 8.8.8.8, 8.8.4.4↵
↵
```

Changing network settings may cause the IP connection to be dropped.

## UI Settings Block

The UI settings block contains the front panel LCD and monitor output settings for the connected Streaming Encoder.

### Block Syntax

```
UI SETTINGS:↵
Available Locales: en_US.UTF-8, zh_CN.UTF-8, ja_JP.UTF-8, ko_KR.UTF-8, es_
ES.UTF-8, de_DE.UTF-8, fr_FR.UTF-8, ru_RU.UTF-8, it_IT.UTF-8, pt_BR.UTF-8,
tr_TR.UTF-8, pl_PL.UTF-8, uk_UA.UTF-8↵
Current Locale: en_US.UTF-8↵
Available Audio Meters: PPM -18dB, PPM -20dB, VU -18dB, VU -20dB↵
Current Audio Meter: PPM -20dB↵
↵
```

## Parameters

| Key                    | Read/Write | Description                                                   | Valid Values                                                    |
|------------------------|------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| Available Locales      | Read only  | The locales available in the Streaming Encoder                | Comma separated list of locales                                 |
| Current Locale         | Read/Write | The current locale for Streaming Encoder                      | Refer to the locales from the Available Locales field           |
| Available Audio Meters | Read only  | The available audio meters supported by the Streaming Encoder | Comma separated list of audio meter types                       |
| Current Audio Meter    | Read/Write | The current audio meter                                       | Refer to the audio meters from the Available Audio Meters field |

## Stream Settings Block

The stream settings block contains the stream configuration for the connected Streaming Encoder.

### Block syntax

This example shows the output for a connected Streaming Encoder. The stream settings prefixed with Current show the active stream settings and are writable. The stream settings prefixed by Available show the available stream settings for the device or platform and are read-only. To view the available servers or quality levels for a stream service, the Current Platform stream setting should be set first.

```
STREAM SETTINGS:↵
Available Video Modes: Auto, 1080p23.98, 1080p24, 1080p25, 1080p29.97,
1080p30, 1080p50, 1080p59.94, 1080p60, 720p25, 720p30, 720p50, 720p60↵
Video Mode: 1080p59.94↵
Current Platform: YouTube↵
Current Server: Primary↵
Current Quality Level: Streaming Medium↵
Stream Key: abc1-def2-ghi3-jkl4-mno5↵
Password: ↵
Current URL: srt://192.168.8.51
Customizable URL: true
Available Default Platforms: YouTube RTMP, YouTube SRT (Beta), Facebook,
Microsoft Teams, Instagram, Twitch, X (Twitter), Restream.IO, Vimeo,
BoxCast, CastR, AfreecaTV, Bilibili, DouYu, Weibo↵
Available Custom Platforms: My Platform↵
Available Servers: Primary, Secondary↵
Available Quality Levels: HyperDeck High, HyperDeck Medium, HyperDeck Low,
Streaming High, Streaming Medium, Streaming Low↵
↵
```



## Parameters

| Key                         | Read/Write | Description                                                                                                  | Valid Values                                                                                                    |
|-----------------------------|------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Available Video Modes       | Read only  | The video modes available in the Streaming Encoder                                                           | Comma separated list of video modes                                                                             |
| Video Mode                  | Read/Write | The current video mode                                                                                       | Refer to the video modes from the <i>Available Video Modes</i> field                                            |
| Current Platform            | Read/Write | The selected streaming platform                                                                              | Refer to the platforms from the <i>Available Default Platforms</i> and <i>Available Custom Platforms</i> fields |
| Current Server              | Read/Write | The current server for the streaming platform                                                                | Server is dependent on selected Current Platform                                                                |
| Current Quality Level       | Read/Write | The current streaming quality level                                                                          | Quality level is dependent on selected Current Platform                                                         |
| Stream Key                  | Read/Write | The stream key for the streaming platform                                                                    | String                                                                                                          |
| Password                    | Read/Write | The passphrase for an encrypted SRT stream                                                                   | String                                                                                                          |
| Current URL                 | Read/Write | The current URL for the streaming platform. This field is writable if <i>Customizable URL</i> field is true. | String                                                                                                          |
| Customizable URL            | Read only  | A boolean specifying whether custom URLs are supported by the streaming platform                             | true - Custom URLs are supported<br>false - Custom URLs are not supported                                       |
| Available Default Platforms | Read only  | The available default streaming platforms                                                                    | Comma separated list of default platforms                                                                       |
| Available Custom Platforms  | Read only  | The custom streaming platforms loaded in the Streaming Encoder                                               | Comma separated list of custom platforms                                                                        |
| Available Servers           | Read only  | The available servers for the selected streaming platform                                                    | Comma separated list of servers                                                                                 |
| Available Quality Levels    | Read only  | The available quality levels for the selected streaming platform                                             | Comma separated list of quality levels                                                                          |

## Changing Stream Settings

The stream settings can be changed by providing a stream settings block. The following is an example of streaming a 1080p59.94 input on Twitch with a medium stream quality.

```

STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
Current Platform: Twitch↵
Current Server: US West: Los Angeles, CA↵
Current Quality Level: Streaming Medium↵
Stream Key: live_123456789_1aB2cD3eF4gH5iJ6kL7mN8oP9qR0sT↵
↵
ACK↵
↵
STREAM SETTINGS:↵
Video Mode: 1080p59.94↵
Current Platform: Twitch↵
Current Server: US West: Los Angeles, CA↵
Current Quality Level: Streaming Medium↵
Stream Key: live_123456789_1aB2cD3eF4gH5iJ6kL7mN8oP9qR0sT↵
↵

```

## Stream XML Block

The stream XML block allows users to configure the Streaming Encoder with a custom configuration file in XML format.

### Block syntax

The following example shows an XML file - Custom.xml has been loaded to configure the stream settings in the Streaming Encoder.

```
STREAM XML:↵
Files: Custom.xml↵
↵
```

### Parameters

| Key    | Read/Write | Description                               | Valid Values                      |
|--------|------------|-------------------------------------------|-----------------------------------|
| Files  | Read/Write | The XML files loaded in Streaming Encoder | Comma separated list of filenames |
| Action | Write only | The stream XML action                     | Remove<br>Remove All              |

### Adding a Stream XML file

An XML file can be loaded onto a Streaming Encoder by sending the stream xml command with a filename, then provide the contents of the XML file. After adding the XML file, the Available Custom Platforms field in the STREAM SETTINGS block will be updated with the new platforms, however the Current items will remain unchanged.

For the Stream XML block to be parsed correctly, any blank lines should be removed from the XML files.

Refer to the `Blackmagic Streaming XML Format` section in this manual for description of the Stream XML file format.

```
STREAM XML Custom.xml:↵
<?xml version="1.0" encoding="UTF-8"?>↵
<streaming>↵
    <service>↵
        <name>My Custom Platform</name>↵
        ...
    </service>↵
</streaming>↵
↵
ACK↵
↵
STREAM XML Custom.xml:↵
<?xml version="1.0" encoding="UTF-8"?>↵
<streaming>↵
    <service>↵
        <name>My Custom Platform</name>↵
        ...
    </service>↵
</streaming>↵
↵
```

```
STREAM XML:↵
Files: Custom.xml↵
↵
STREAM SETTINGS:↵
Available Custom Platforms: My Custom Platform↵
↵
```

### Removing a Stream XML file

An XML file can be removed from the Streaming Encoder by sending the stream xml command with the remove action.

```
STREAM XML:↵
Action: Remove↵
Files: Custom.xml↵
↵
ACK↵
↵
STREAM XML:↵
Files:↵
↵
STREAM SETTINGS:↵
Available Custom Platforms:↵
↵
```

### Removing all Stream XML files

All XML files can be removed from the Streaming Encoder by sending the stream xml command with the remove all action. In the example, following the remove all action, the loaded stream XML files and available custom platforms are both displayed as empty.

```
STREAM XML:↵
Action: Remove All↵
↵
ACK↵
↵
STREAM XML:↵
Files: ↵
↵
STREAM SETTINGS:↵
Available Custom Platforms:↵
↵
```

## Stream State Block

The stream state block provides the streaming status of the Streaming Encoder.

The Streaming Encoder server will send a stream state block update whenever there is a change to the Status field. Due to frequency of changes to Duration, Bitrate and Cache Used fields, these fields need to be polled by the client by requesting a Stream State block.

### Block syntax

```
STREAM STATE:↵
Status: Idle↵
Bitrate: 161672↵
Duration: 00:00:00:00↵
Cache Used: 0↵
↵
```

### Parameters

| Key        | Read/Write | Description                                                                       | Valid Values                                   |
|------------|------------|-----------------------------------------------------------------------------------|------------------------------------------------|
| Status     | Read only  | The stream state of the Streaming Encoder, updated when the stream status changes | Idle<br>Connecting<br>Streaming<br>Interrupted |
| Action     | Write only | The Streaming Encoder stream state action.                                        | Start<br>Stop                                  |
| Duration   | Read only  | The duration of the active stream                                                 | String in format of DD:HH:MM:SS                |
| Bitrate    | Read only  | The bitrate of the active stream                                                  | Integer bits per second                        |
| Cache Used | Read only  | The current usage of the streaming cache                                          | Integer as a percentage                        |

### Starting Stream

The stream is started by providing a stream state block with start action.

```
STREAM STATE:↵
Action: Start↵
↵
ACK↵
↵
STREAM STATE:↵
Status: Connecting↵
↵
STREAM STATE:↵
Status: Streaming↵
↵
```

### Stopping stream

The stream is stopped by providing a Stream State block with stop action.

```
STREAM STATE:↵
Action: Stop↵
↵
ACK↵
↵
STREAM STATE:↵
Status: Idle↵
↵
```

### Audio Settings Block

The Audio Settings block contains the audio configuration for the connected Streaming Encoder.

#### Block syntax

This example shows the output for a connected Streaming Encoder. The stream settings prefixed with Current show the active audio settings and are writable. The stream settings prefixed by Available show the available audio settings for the device or platform and are read-only.

```
AUDIO SETTINGS:↵
Current Monitor Out Audio Source: Auto↵
Available Monitor Out Audio Sources: Auto, SDI In, Remote Source↵
↵
```

#### Parameters

| Key                                 | Read/Write | Description                                                          | Valid Values                                                                         |
|-------------------------------------|------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Current Monitor Out Audio Source    | Read/Write | The current audio source on the monitor output                       | Refer to the audio sources from the <i>Available Monitor Out Audio Sources</i> field |
| Available Monitor Out Audio Sources | Read only  | The available audio sources that can be routed to the monitor output | Comma separated list of audio sources                                                |

### Changing Audio Settings

The audio settings can be changed by providing an audio settings block. The following is an example of setting the monitor output audio source to remote.

```
AUDIO SETTINGS:↵
Current Monitor Out Audio Source: Remote Source↵
↵
ACK↵
↵
AUDIO SETTINGS:↵
Current Monitor Out Audio Source: Remote Source↵
↵
```

## Shutdown Block

The Shutdown block provides power control of the Streaming Encoder. The Shutdown block is write-only and not presented in the preamble.

### Parameters

| Key    | Read/Write | Description                            | Valid Values            |
|--------|------------|----------------------------------------|-------------------------|
| Action | Write only | The Streaming Encoder shutdown action. | Reboot<br>Factory Reset |

### Reboot

The Streaming Encoder can be rebooted by providing a Shutdown block with reboot action.

SHUTDOWN:↵

Action: Reboot↵

↵

ACK↵

↵

On reboot action, the Streaming Encoder server will be stopped and clients will be disconnected.

### Factory Reset

The Streaming Encoder can be factory reset by providing a Shutdown block with factory reset action. On factory reset action, all settings are set to factory defaults.

SHUTDOWN:↵

Action: Factory Reset↵

↵

ACK↵

↵